FUNCTIONAL PROTOTYPE

SMMG

Healthcare Appointment Management System with an AI-Powered Health and First-Aid Assistant

https://smmg.ruzzgd.dev

Prototype Explanation and Page Guide

Prepared for client review and system evaluation

August 2026

1. Purpose of This Prototype

SMMG is a working prototype of a web-based healthcare appointment management system. It demonstrates how patients, doctors, clinic staff, and administrators interact with one shared platform.

The prototype allows the client and project stakeholders to review the design, navigation, workflows, and core functions before the system is considered ready for final production use.

> **Prototype status:** The client may explore and test the system now, but only test or demonstration data should be entered. Real patient records, confidential medical information, and live clinic operations should not yet be processed through this staging version.

2. System Overview

The system is organized around distinct user roles, each with tailored interfaces and permissions designed to streamline their specific tasks.

USER ROLE	PRIMARY RESPONSIBILITIES & CAPABILITIES
Public (Guest)	Explore clinic services, browse doctor directories by specialty, read legal policies, and register for a new patient account.
Patient	Book appointments, manage upcoming and past consultations, update personal profiles, and interact with the AI assistant for inquiries.
Doctor	Review daily consultation schedules, mark appointments as completed, update specialty profiles, and view personal performance reports.
Staff	Monitor the daily clinic queue, confirm pending appointments, update appointment statuses, and assist with patient flow management.
Administrator	Manage system configurations (services, specialties, users), generate comprehensive analytical reports, oversee doctor schedules, and audit AI conversations.

3. Main Prototype Workflow

The following sequence illustrates the complete lifecycle of a standard appointment within the system:

1. **Registration & Discovery** — A guest registers for an account, verifies their email via a 6-digit code, and browses the directory of doctors or services.
2. **Booking** — The patient selects a doctor, chooses an available date and time slot, specifies a reason for the visit, and confirms the booking.
3. **Review & Confirmation** — A Staff member or Administrator reviews the pending appointment on their dashboard and updates the status to "Confirmed".
4. **Consultation** — On the scheduled day, the Doctor views the appointment on their dashboard, conducts the consultation, and marks the appointment as "Completed".
5. **Analytics** — Administrators generate reports to analyze consultation volume, service distribution, and clinic performance based on the completed appointments.

4. Public and Authentication Pages

These pages form the entry point for all users, providing essential clinic information and secure access to the system.

ROUTE PATH	PURPOSE & KEY FEATURES
`/`	Landing page featuring a hero section, benefits grid, a 4-step "How it works" guide, featured services and doctors, and promotional AI assistant content. Automatically redirects authenticated users to their respective dashboards.
`/login`	Authenticates users via email and password, handling unverified email redirects and displaying verification success banners. Redirects users to their role-specific dashboards upon success.
`/register`	Account creation form requiring name, email, and password. Features client-side validation (8+ characters, password matching) and redirects to email verification upon submission.

ROUTE PATH	PURPOSE & KEY FEATURES
`/verify-email`	Processes a 6-digit verification code with masked email display. Includes a 30-second cooldown for resending codes and handles context-aware redirects (e.g., login return).
`/doctors`	Responsive 3-column directory of available doctors, featuring a specialty filter dropdown and profile cards displaying avatars, titles, specialties, bios, and contact info.
`/doctors/:id`	Detailed doctor profile integrating a full appointment booking widget. Allows patients to select dates (preventing past dates), choose services, pick available time slots from an API, and confirm bookings with a reason/symptoms input. Generates a reference number upon success.
`/services`	Displays a grid of clinic services with descriptive images (or fallback SVGs), specialty badges, and detailed descriptions.
`/specialties`	Categorized grid of medical specialties featuring icons, descriptions, and direct links to view doctors filtered by the selected specialty.
`/privacy`	Comprehensive privacy policy document with a sticky sidebar Table of Contents navigating through 7 distinct sections.
`/terms`	Detailed terms of service agreement equipped with a sidebar Table of Contents covering 9 sections.

5. Patient Pages

Designed for individuals seeking medical care, these pages require Patient role authentication.

ROUTE PATH	PURPOSE & KEY FEATURES
`/patient/dashboard`	Central hub displaying a welcome banner, account status metrics, details of the next upcoming appointment, and a quick-action grid for accessing the AI Assistant, browsing doctors, viewing services, and managing appointments.
`/patient/appointments`	Comprehensive appointment management using tabbed navigation (Upcoming vs. History & Cancelled). Features detailed appointment cards, a cancellation modal requiring a reason (enforcing a 24-hour policy), and a rescheduling tool with date/slot selection.
`/patient/profile`	Personal information management allowing avatar uploads, name and demographic updates (DOB, gender), address editing, and Philippine mobile number validation. Includes a secure Change Password component.

AI Assistant (Available to All Authenticated Users)

The AI Assistant is accessible to all authenticated users (Patients, Doctors, Staff, and Administrators) for general health and first-aid information.

ROUTE PATH	PURPOSE & KEY FEATURES
`/ai-assistant`	Interactive chat interface featuring a sidebar for managing sessions (create, switch, delete). Includes distinct message styling (user vs. AI), typing indicators, markdown rendering for rich text, and a prominent Philippine 911 emergency disclaimer.

> **AI assistant limitation:** The assistant is an informational prototype and is not a replacement for a physician, diagnosis, emergency service, or professional medical advice.

6. Doctor and Staff Pages

These interfaces provide medical professionals and clinic staff with the tools necessary to manage their daily schedules and patient interactions.

Doctor Pages

ROUTE PATH	PURPOSE & KEY FEATURES
`/doctor/dashboard`	Overview displaying a welcome banner and metric cards (today's consultations, upcoming, active schedules). Highlights today's appointments in a table with quick actions to mark consultations as completed.
`/doctor/appointments`	Detailed listing of all scheduled consultations with date and status filtering. Displays patient information, required services, and allows doctors to manage appointment statuses directly.
`/doctor/profile`	Professional profile management including avatar upload, credential updates, specialty assignment, bio editing, and secure password changes.
`/doctor/reports`	Performance analytics featuring date range filtering, key metric cards (total, completed, completion rate, cancelled), CSS-based bar charts illustrating consultation trends, and a service breakdown table.

Staff Pages

ROUTE PATH	PURPOSE & KEY FEATURES
`/staff/dashboard`	Operational hub featuring 5 metric cards tracking daily totals and statuses. Displays today's appointment table with essential quick actions to confirm pending requests or complete finalized consultations.

> **Note:** Staff members also have access to `/admin/appointments` and `/admin/reports` for broader operational oversight.

7. Administrator Pages

Dedicated to system configuration, user management, and high-level analytics, requiring Administrator privileges.

ROUTE PATH	PURPOSE & KEY FEATURES
`/admin/dashboard`	High-level oversight featuring a System Metrics Overview (appointment stats, CSS donut charts for status distribution, horizontal bar charts for user roles) and a User Accounts management tab with role filtering and account activation toggles.
`/admin`	Central configuration portal with tabs for managing Doctors, Staff, Specialties, and Services. Supports adding new entries via comprehensive modals, including image upload widgets for services.
`/admin/appointments`	Master view of all system appointments featuring multi-criteria filtering (doctor, status, date) and full status management capabilities.
`/admin/schedules`	Configuration of doctor availability. Allows administrators to select doctors, view weekly schedules, and add or modify active time slots including defining slot durations and time ranges.
`/admin/reports`	Advanced analytics suite with CSV export functionality and date range presets. Features 4 analytical tabs exploring Volume & Status, Doctor Performance, Services & Specialties distribution, and Peak Times & Demographics (utilizing progress bars, data grids, and gender/age group visualizations).
`/admin/ai-monitoring`	Compliance and safety tool presenting a split layout for searching and reviewing read-only AI conversation logs, prefaced with a privacy warning.

Shared Pages

ROUTE PATH	PURPOSE & KEY FEATURES
`/account/profile`	Standardized profile management interface utilized by Staff and Administrators for updating avatars, names, and passwords.

8. Prototype Features Available for Evaluation

The prototype includes several advanced technical implementations and UI components designed to enhance the user experience without relying on heavy external dependencies.

- ✓ **In-App Notification System** — Real-time alerts featuring unread badges and state-tracking for read/unread status.
- ✓ **Secure Password Recovery** — A robust two-step email-verified password change flow (request code → validate code and set new password).
- ✓ **Localized Data Validation** — Strict validation and normalization for Philippine mobile numbers across all input forms.
- ✓ **Client-Side Data Export** — Functionality for doctors and administrators to generate and download CSV reports directly from the browser.
- ✓ **Native CSS Visualizations** — Implementation of donut charts, horizontal/vertical bar charts, and progress bars using pure CSS, eliminating the need for external charting libraries.
- ✓ **Rich Media Handling** — Image upload widgets for services including preview, replacement, and removal capabilities.
- ✓ **Intelligent Routing** — Role-based automatic redirection ensuring users are directed to their appropriate dashboards.
- ✓ **Dynamic UI Components** — An adaptive public header that transitions from transparent on the homepage to solid on content pages, improving readability.
- ✓ **Safety-Conscious AI Integration** — The AI assistant incorporates explicit emergency contact disclaimers to ensure safe usage in health-related contexts.
- ✓ **Integrated Booking Flow** — A seamless appointment booking experience directly embedded within doctor profiles, progressing logically from date selection to service choosing, slot picking, and confirmation.

9. Suggested Client Evaluation

The client should evaluate the prototype using prepared demonstration accounts and test data. The following scenarios are recommended:

1. Browse doctors, specialties, and services as a public visitor.
2. Register and verify a test patient account.
3. Complete the patient profile and upload a test avatar.
4. Create an appointment from a doctor's profile page and confirm it appears in the patient account.
5. Review the appointment through a doctor, staff, or administrator account.
6. Update the appointment status and confirm the patient sees the change.
7. Review the administration dashboard and reporting screens.
8. Test the AI assistant using general, non-confidential questions.
9. Check the system on both desktop and mobile devices.

10. Feedback and Approval Checklist

Please use this checklist to evaluate the prototype and note any required adjustments.

REVIEW AREA	QUESTIONS FOR THE CLIENT
Workflow	Does the appointment process match the clinic's expected procedure?
Roles and permissions	Can each user access the functions they need without unnecessary options?
Information	Are all required fields, statuses, services, and reports included?
Interface	Are labels, navigation, page layout, and mobile behavior clear?
Notifications	Are confirmations, errors, and status messages understandable?
AI Assistant	Are the assistant responses relevant, and are disclaimers visible?
Final requirements	What must be changed or added before production approval?

> **Recommended approval statement:** "The client has reviewed the SMMG functional prototype and has provided feedback on the demonstrated pages, workflows, and features. Approval of the prototype confirms the intended direction of the system but does not yet represent production acceptance."

11. Next Steps Before Live Use

- ☐ Collect and document client feedback.
- ☐ Implement and verify the approved revisions.
- ☐ Complete functional, usability, security, and data-handling tests.
- ☐ Prepare production configuration, credentials, storage, backups, and monitoring.
- ☐ Create authorized production user accounts and provide user training.
- ☐ Obtain final acceptance before processing real clinic or patient information.

This document explains the current SMMG functional prototype for demonstration and client review. It is not a medical, legal, privacy-compliance, or production-readiness certification.